

ORIGINAL ARTICLE

Comparative efficacy of depotphoresis and diode laser for reduction of microbial load and postoperative pain, and healing of periapical lesions: a randomized clinical trial

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Introduction

The main goal of endodontic treatment in teeth with necrotic pulp and chronic periapical lesions is to create a germ-free environment in the root canal system, and the adjacent periodontal tissue.

Endodontists generally have two main goals in root canal treatment. The short-term goal is to prevent/control pain, and the long-term goal is to achieve clinical and radiographic healing. To achieve this goal, the microbial load in the root canal system should be minimized (1). There are two main obstacles against achieving this goal namely the complex root canal anatomy and the unique properties of the bacterial flora in the root canal system. The microorganisms in the root canal system are inaccessible in some cases despite the use of advanced endodontic instruments (2). The residual bacteria in the root canal system can proliferate between the treatment sessions and reach their baseline preoperative count (3).

The treatment success has several criteria such as resolution of symptoms, periapical healing on radiographs, or histological evidence of healing. A number of factors affect the success of endodontic treatment such as the severity of condition, the instrumentation technique, and the technology used for treatment (4).

In the recent years, use of laser in endodontics has gained increasing popularity, yielding acceptable results as an adjunct for disinfection of the root canal system and reduction of bacterial load (5, 6). Several studies have confirmed the optimal efficacy of laser therapy for reduction of post-endodontic pain as well (7, 8).

Depotphoresis is another method of root canal treatment in which, the root canal is filled with Cupral (a copper-calcium-hydroxide based compound) and the electric current is guided into the root canal system with a fine probe. This technique dissolves the contents of the root canal by alkaline proteolysis, eliminates the oligopeptides formed in the root canal system, causes proteolysis, and eliminates the acidic bio-

logical remnants in the periapical region. Cupral penetrates into the dentinal tubules and disinfects them, and induces osteogenesis in the periapical region (9). Addition of copper ions to calcium hydroxide plays an important role in the process of tissue regeneration, and also exerts anti-inflammatory and disinfecting effects (10). Moreover, the electrically activated copper ions used in combination with calcium hydroxide exert significantly faster antibacterial effects in dentinal tubules compared with pure calcium hydroxide without electrical activation (11). However, it should be noted that studies on this topic are limited (9).

Cone-beam computed tomographic (CBCT) imaging is often used for diagnosis and treatment planning in endodontics. Also, CBCT enables the measurement of periapical lesions in three orthogonal planes (12). Moreover, evidence shows the superiority of CBCT for monitoring of the healing course of periapical lesions compared with periapical radiography (13). The novel root canal disinfection techniques, such as DL and depotphoresis, have improved the outcome of endodontic treatment.

Yalgi et al. (9) found that depotphoresis is effective in reducing postoperative pain and in the treatment of chronic apical periodontitis.

Some authors (14, 15) proposed that DL may be a successful adjunct to endodontic treatment of necrotic cases with chronic periapical lesions in terms of postoperative pain and root canal disinfection.

This study aimed to compare the effects of depotphoresis and DL on microbial count, postoperative pain, and healing of periapical lesions after 6 months by using CBCT.

Materials and Methods

The study population comprise of 90 patients presenting to the Endodontics Department of School of Dentistry, Zahedan University of Medical Sciences who required endodontic treatment of teeth with pulp necrosis associated with chronic apical periodontitis. The patients were



between 18 to 60 years, and the diagnosis of pulp necrosis with chronic apical periodontitis was confirmed by radiographic examination and negative response to cold test and electric pulp test, and no bleeding when exposing the pulp.

The protocol of this prospective, three-arm, parallel-group, single-blind, single-center, randomized clinical trial and the informed consent form were approved by the Research Ethics Committee of Zahedan University of Medical Science (IR.ZAUMS.REC.1397.278).

The study protocol was registered at <https://en.irct.ir/trial/3284> (Iranian Registry of Clinical Trials identification number IRCT20180718040517N1).

This study was conducted and reported in accordance with the consolidated standards of reporting trials (CONSORT) (19). All enrolled patients signed informed consent forms after the nature of the study, its objectives, procedures, benefits and potential risks were explained to them.

Sample size calculation

A pilot study was designed to determine the size of the study population. Data obtained from the pilot study indicated that the sample size for each group should be a minimum of 25. Considering the possibility of dropouts, 30 patients were allocated to each group (a total of 90) assuming the effect size of 0.82, power of 80%, and 0.05 level of significance. The patients were enrolled from the outpatient clinic of the Department of Endodontics, Faculty of Dentistry between December 2018 and June 2020. One postgraduate student of endodontics performed the procedures.

Inclusion and exclusion criteria

The patients were between 18 to 60 years, and the diagnosis of pulp necrosis was confirmed by negative response to cold test and electric pulp test, and no bleeding when exposing the pulp. Diabetic patients, immunocompromised patients, pregnant women, and those with a history of analgesic intake in the past 3 days or antibiotic intake in the past month were excluded. Teeth with periodontal disease, or previ-

ous access cavity and those that could not be isolated by rubber dam were excluded as well.

Randomization, allocation concealment and masking of examiners

The patients were allocated to the three study arms with simple randomization. Randomization was performed online (endo.dent.zaums.ac.ir) right before the treatment onset. After recording the patients' demographics, eligible patients were automatically randomized into the three study groups of depotphoresis, diode laser, and control by using a computer-generated system. All the allocated patients received a unique patient identification code before starting the treatment. In the event of withdrawal from the study, the randomization code was not reused. Patient blinding was implemented. In Control group, in 15 patients, the fiber optic was placed inside the root canals without activation and in 15 other patients, red clip of depotphoresis was placed on the patient's cheek and blue clip was placed above the tooth's root canal without any current.

In the DL group, without any current, Red and blue clips, were placed inside the patient's mouth, as mentioned earlier. All patients and the operator wore protective glasses during all stages of the operation in all groups.

In the depotphoresis group, a mock application of laser was made with the power off.

Interventions

In the first session, the level of pain of patients was recorded before treatment using a visual analog scale (VAS). After dental flossing and dental plaque removal by pumice paste, rubber dam isolation was performed, and the tooth and the treatment area were disinfected with 30% H₂O₂ (v/v) for 30 s followed by 2.5% NaOCl. The access cavity was prepared with sterile high-speed bur (Tizcavan, Tehran, Iran) without water spray. After completion of access cavity preparation and caries removal (prior to exposure of the pulp chamber), the tooth, the clamp,

and the rubber dam were disinfected again with 2.5% NaOCl; 5% sodium thiosulfate was then used to neutralize sodium hypochlorite. Dental pulp was then exposed. Next, three sterile paper points were placed in the canal for 1 min to collect the first microbial sample (prior to root canal preparation). The collected sample (S1) was transferred into a tube containing HBI buffer and frozen at -20 °C.

A sterile #15H-file (Dentsply/Maillefer, Ballaigues, Switzerland) was then introduced into the canal, and the working length was determined by an apex locator (ROOT ZX; J. Morita, Tokyo, Japan) and confirmed radiographically. The narrow root canals were prepared with 25/8% Reciproc file (VDW, Munich, Germany) while the wide canals were prepared with 40/6% Reciproc file along with Reciproc Silver electric motor. Also, 2.5% NaOCl and 17% EDTA (Aria Dent, Tehran, Iran) were used for root canal irrigation. After chemomechanical preparation of the root canals, sterile saline was injected into the canals by a 27-gauge needle to a level below the orifice. Next, #30 Hedstrom file (Dentsply, Maillefer, Ballaigues, Switzerland) was used for filing and agitation of the canal content. This was done to enhance the release of dentin chips into the saline. Subsequently, three sterile paper points were introduced into the canal one by one and remained there for 60 s to collect the second sample (S2). Under aseptic conditions, the collected sample was transferred into a tube containing HBI buffer by a sterile hemostat and frozen at -20 °C. It was transferred to the lab on the same day.

Depotphoresis (Humamchemic GmbH, Alfeld, Germany) group: Cupral paste was prepared according to the manufacturer's instructions (to prepare Cupral solution, 1 unit of Cupral was mixed with 9 units of calcium hydroxide) and delivered into the coronal third of the canal by a Lentulo spiral. The positive electrode was connected to the patient's buccal mucosa by a clamp. The negative electrode was placed in the pulp chamber and the electric current was initiated and gradually increased (the current suggested by the manufacturer is 0.8-1.5 mA/5 min). The tooth was

temporarily restored with Cavit (Golchay, Tehran, Iran) while calcium hydroxide Cupral remained in the canal as medication.

Laser group

After root canal preparation, it was rinsed with saline and cleaned with paper points. Next, 940 nm diode laser (EPIC X, BIO-LASE, Irvine, CA, USA) hand-piece with 200 µm tip diameter (EZTIP400&300) was introduced into the canal to 1 mm from the apex and was then pulled back with a circular motion at a speed of 2 mm/s towards the orifice. After 20 s of laser irradiation in pulse-mode, dental substrate was allowed to rest for 10 s. Laser irradiation was repeated 3 times. The laser parameters included $\lambda=940$ nm and 2 W power. The second samples were then collected (S2). The root canals were filled with a mixture of calcium hydroxide and saline, delivered into the canal by a Lentulo spiral (Dentsply Maillefer) as medicament, and temporarily restored with Cavit (Golchay, Tehran, Iran) between the treatment sessions.

Control group

After root canal disinfection and instrumentation, a mixture of calcium hydroxide and saline was delivered into the canal by a Lentulo spiral as medicament and temporarily restored with Cavit (Golchay, Tehran, Iran) between the treatment sessions.

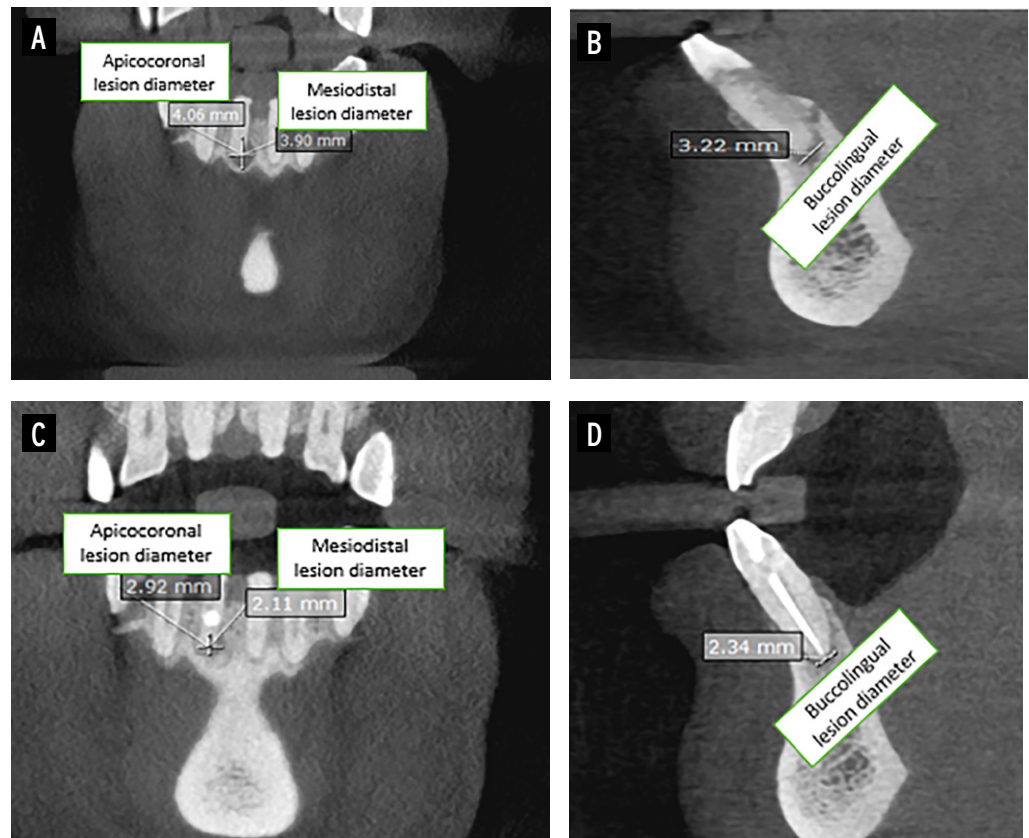
Post-endodontic pain assessment

After completion of the first treatment session, the patients received a VAS form (graded from 0 to 10). They were instructed on how to fill out the VAS form and were requested to express their level of pain by selecting the score that best described their pain severity at 6, 12, 24, 48 and 72 h, post-treatment. A recall visit was also scheduled for them 7 days after treatment. Independent examiners who were blinded to the group allocation of patients evaluated the VAS forms.

Second treatment session

After removing the intracanal medica-

Figure 1
A, B Measurement of the preoperative diameter of the lesion and **C, D** diameter of the lesion after 6 months.



ments (calcium Cupral in depotphoresis group and calcium hydroxide in laser and control groups), the third sampling (S3) was performed as explained earlier. The teeth were then obturated and filled with composite resin. All endodontic treatments were performed by a post-graduate student of endodontics within 8 months.

Microbial analysis

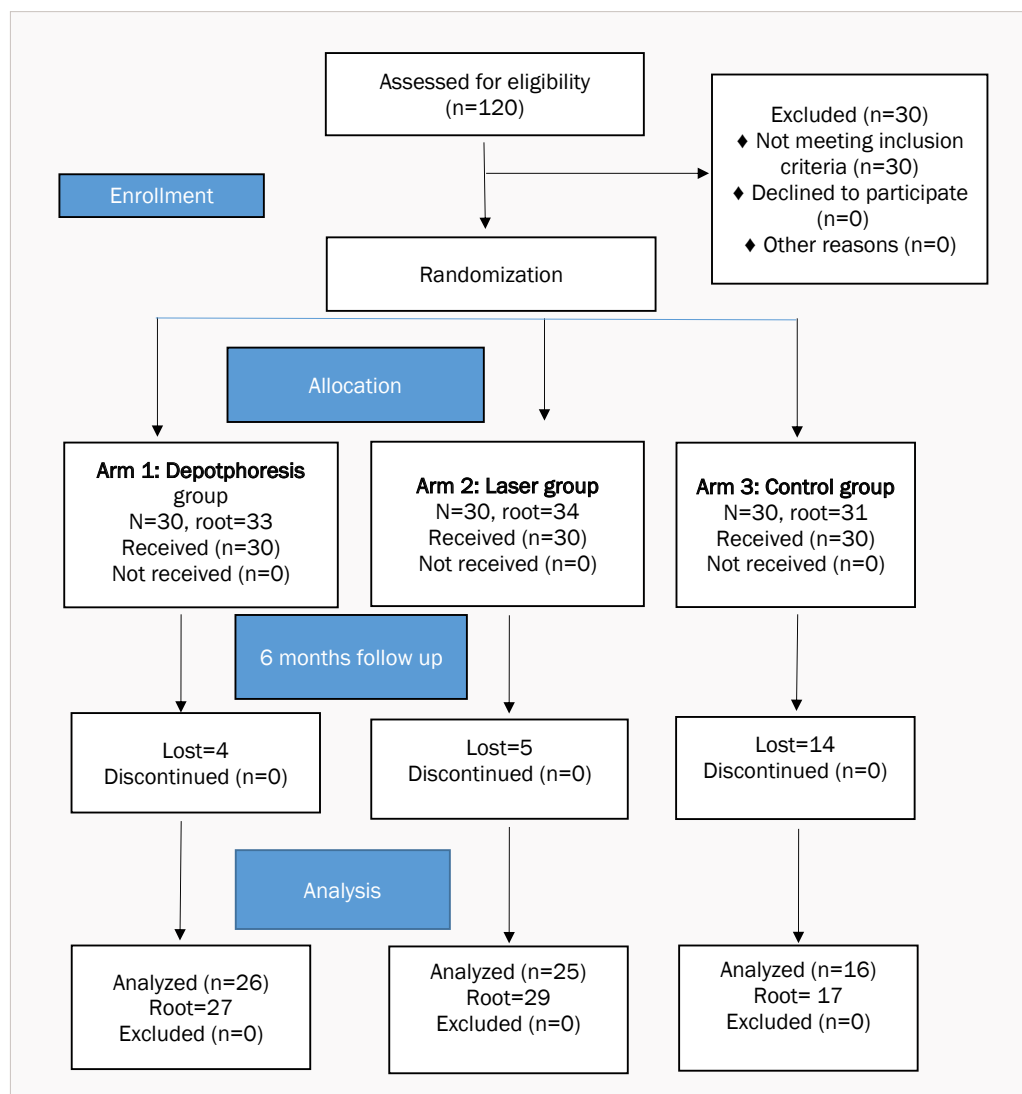
The microbiological assessment was conducted by the “Colony Forming Unit Counting” method. Samples were transported to the laboratory within two hours after collection for microbiologic assessment. Samples in RTF vials were mixed well with a vortex for 30 seconds to facilitate bacterial dispersion. 10-fold serial dilutions were made in Phosphate Buffer Saline (PBS) solution. 10 ml of the sample was transferred to a sterile tube. 10 tubes containing 9 ml sterile saline were placed in a tube holder. An aliquot of 1 ml from the original sample was then transferred into the first tube containing 9ml of sterile saline. This process was repeated for each

of the remaining 9 tubes. At the end of the 10-fold dilutions, 10 tubes covering the dilution range of $1:10^{-1}$ through to $1:10^{-10}$ had been made. For each tube, 1ml of the diluted sample was transferred to BHI agar plates and cultured under aseptic conditions. The plates were incubated at 37°C for 24 hours for the bacteria to grow. The number of bacterial colonies in each plate was visually quantified and reported as colony forming units per milliliter (CFU/ml). The number of bacteria (CFU) per milliliter of sample was calculated by dividing the number of colonies by the dilution factor. At last the outer numbers were thrown out and mean of the rest of numbers were calculated.

Radiographic technique

The healing of periapical lesions was evaluated on CBCT scans. Prior to the treatment onset, the respective tooth underwent CBCT. All primary and follow-up CBCT scans were performed by Acteon CBCT scanner (Acteon Group, Norwich, United Kingdom). All CBCT scans were obtained

Figure 2
Consort flowchart
of the study.



with 40 mm³ field of view and 0.2 µm voxel size with the exposure settings of 8 mA, 85 kVp and 10 s exposure time. All CBCT images were analyzed using OnDemand 3D software (CyberMed, Seoul, Republic of Korea).

Radiographic assessment

The following parameters were evaluated. Diameter of lesions: the maximum diameter of periapical lesions was measured in three directions of apicocoronal (Ly), mesiodistal (Lx) and buccolingual (Lz) parallel to the standard axis according to Kim et al (16) (Figure 1) If the lesion did not have a bony margin (for example, if the cortical plate or sinus wall had been per-

forated), the examiner would be asked to estimate the size of the lesion based on the visible bone margin at each side of the bone defect. The radiographic findings were analyzed according to Patel et al. (17) with some differences. Since the present study focused on root canal treatment of the teeth with chronic apical periodontitis, the codes 1 and 6 from the study by Patel et al. (17) were not applicable to our study. The CBCT images were evaluated by two endodontists, and in case of disagreement, a third endodontist was asked to cast the final judgment.



Table 1

access cavity preparation (S1) after



of the lesions in the control group showed cant reduction in the bacterial count was









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